Marcos Medina

Telecommunication Engineer · Applied AI and automation

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Castro Urdiales, Spain (Bilbao area)

PORTFOLIO Each project in depth: architecture, decisions and results. **marcosmedina.online →**

PROFILE

Telecommunication Engineer (qualifying MSc, UPV/EHU) specialising in applied AI and automation. I design and ship complete systems on real telephony and networks: AI voice agents in production at many companies and the operations console a technical team works from.

EXPERIENCE

Ausarta · Bilbao, Spain — Sep 2025 – present

AI Development & Automation Engineer — Mar 2026 – present

- **AI voice agents.** A multi-tenant platform of AI phone receptionists, in production at many companies: they answer, put callers through to the right person and take messages at any hour.
- **My own orchestration layer.** It sits between the voice platform and the language model and supervises it in real time: it enforces the steps of each process, applies deterministic rules and fails over between providers.
- **Evaluation and quality.** A test bench with real calls, a simulated caller and a blind automated judge: my architecture beat the platform's native options on quality and speed. Automated tests gate every deployment.
- **NOP, a network operations platform.** A console built from scratch that brings routers, 4G backup, phone systems, phones and tickets into one record per customer: it replaces seven tools and the whole technical team uses it daily.

Telecommunications Engineer · Networks and IP telephony — Sep 2025 – Feb 2026

- Networks and IP telephony for business customers: MikroTik routers, VPN, fibre and 4G backup, Yeastar and 3CX phone systems, SIP trunks and IP phones. Incident diagnosis and resolution.

Elecnor Sistemas · Bilbao, Spain — Jun 2025 – Sep 2025

Intern · Assistant Project Manager, security systems

- CCTV, access control and intrusion projects: documentation, cost tracking and field coordination.

EDUCATION

MSc in Telecommunication Engineering · University of the Basque Country (UPV/EHU) — 2023 – 2025

Qualifies for Spain's regulated engineering profession. Erasmus+ at Jan Długosz University, Poland (2024–2025).

Thesis: 3D localisation of partial discharges with convolutional neural networks (CNN) and iterative optimisation; dataset from electromagnetic simulation (CST Studio Suite) and experimental validation with SDR.

Bachelor's in Telecommunication Technology Engineering · UPV/EHU — 2018 – 2023

Telecommunication Systems specialisation. **Final project:** RF interference caused by high-voltage installations.

SKILLS

AI	Real-time LLM orchestration, STT/TTS, prompt and tool design, automated evaluation
Development	Coding agents and multi-agent workflows, architecture, APIs and webhooks, testing
Networks, VoIP	Yeastar, 3CX, VoIP/SIP, MikroTik RouterOS (BGP, MPLS, OSPF), VPN, Fortinet, Zabbix, GLPI
Signal and ML	CNNs, electromagnetic simulation (CST Studio Suite), SDR, signal processing
Certifications	MikroTik MTCINE (2025), MTCRE and MTCNA
Languages	Spanish (native) · English B2 (Cambridge B2 First, 2018)